

CYBR 343 – W6Lab:

Lab Overview:

In this lab, you will analyze a Windows RAM image to identify and extract forensic artifacts using Volatility 3. Memory forensics allows investigators to uncover evidence that may not exist on disk, including running processes, logged-in users, network connections, and environment variables. You will use Volatility 3 to examine the provided RAM dump and answer a series of investigative questions. If you are using the provided virtual machine, Volatility 3 is already installed. If you are using macOS, you must have Volatility 3 installed as instructed earlier this week.

Learning Objectives

By completing this lab, students will be able to:

1. Analyze a Windows RAM image using Volatility 3.
2. Identify system, user, process, and network artifacts from volatile memory.
3. Use forensic commands to extract and interpret memory-based evidence.
4. Support forensic findings with proper documentation and screenshots.

Required Tools

- Volatility 3
- Provided Windows RAM image

Important Instructions

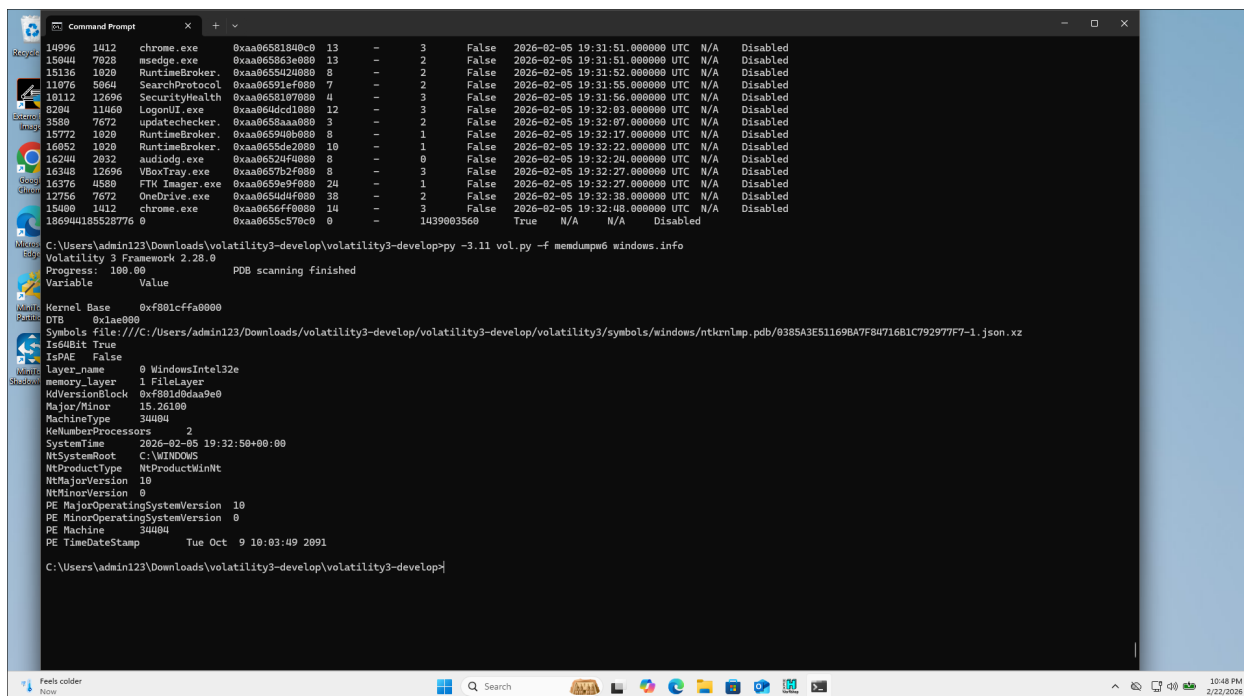
- You must use Volatility 3 for all analysis.
- Answer all questions using evidence extracted from the RAM image.
- Every answer must be supported by screenshots clearly showing:
 - The Volatility command used
 - The corresponding output
- Screenshots must be readable and clearly labeled.

Lab Questions

Q1. System Boot Time

- What time was this system booted?

7:32 PM on 02-05-2026

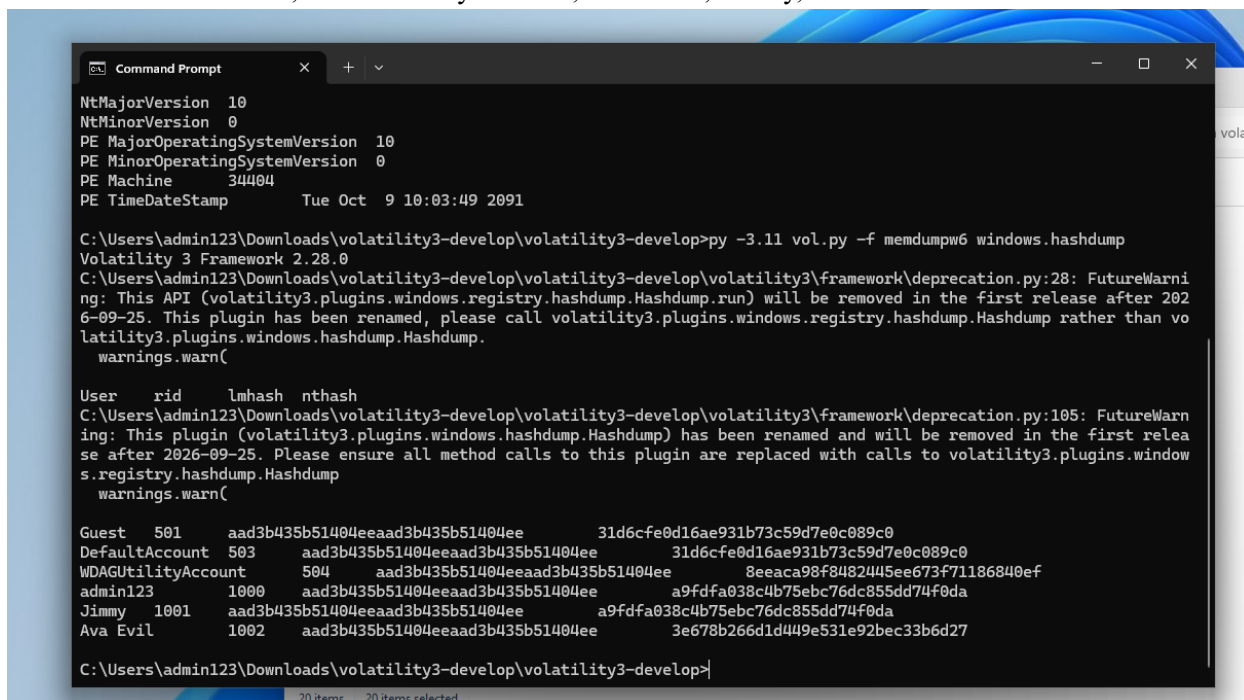


```
Command Prompt
C:\Users\admin123\Downloads\volatility3-develop\volatility3-develop>py -3.11 vol.py -f memdump windows.info
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
Variable Value
Kernel Base 0xf801cfa0000
DTB 0x1ae000
Symbols file:///C:/Users/admin123/Downloads/volatility3-develop/volatility3/symbols/windows/ntkrnlmp.pdb/0305A3E51169BA7F84716B1C792977F7-1.json.xz
Is64Bit True
IsPAE False
Layer_name 0 WindowsIntel32e
memory_layer 1 FileLayer
KdVersionBlock 0xf801d0da9e0
Major/Minor 15.26100
MachineType 34404
NumberOfProcessors 2
SystemTime 2026-02-05 19:32:50+00:00
NTSystemRoot C:\WINDOWS
NTProductType NTProductWinnt
NTMajorVersion 10
NTMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Tue Oct 9 10:03:49 2091
C:\Users\admin123\Downloads\volatility3-develop\volatility3-develop>
```

Q2. User Accounts

- How many user accounts exist on this system?

6 user accounts were identified using Volatility3's hashdump command. The accounts are Guest, DefaultAccount, WDAGUtilityAccount, admin123, Jimmy, & Ava Evil



```
Command Prompt
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Tue Oct 9 10:03:49 2091

C:\Users\admin123\Downloads\volatility3-develop\volatility3-develop>py -3.11 vol.py -f memdump windows.hashdump
Volatility 3 Framework 2.28.0
C:\Users\admin123\Downloads\volatility3-develop\volatility3-develop\volatility3\framework\deprecation.py:28: FutureWarning: This API (volatility3.plugins.windows.registry.hashdump.Hashdump.run) will be removed in the first release after 2026-09-25. This plugin has been renamed, please call volatility3.plugins.windows.registry.hashdump.Hashdump rather than volatility3.plugins.windows.hashdump.Hashdump.
  warnings.warn(

User rid lmhash nthash
C:\Users\admin123\Downloads\volatility3-develop\volatility3-develop\volatility3\framework\deprecation.py:105: FutureWarning: This plugin (volatility3.plugins.windows.hashdump.Hashdump) has been renamed and will be removed in the first release after 2026-09-25. Please ensure all method calls to this plugin are replaced with calls to volatility3.plugins.window
s.registry.hashdump.Hashdump
  warnings.warn(

Guest 501 aad3b435b51404eeaad3b435b51404ee 31d6cfe0d16ae931b73c59d7e0c089c0
DefaultAccount 503 aad3b435b51404eeaad3b435b51404ee 31d6cfe0d16ae931b73c59d7e0c089c0
WDAGUtilityAccount 504 aad3b435b51404eeaad3b435b51404ee 8eeaca98f8482445ee673f71186840ef
admin123 1000 aad3b435b51404eeaad3b435b51404ee a9fdfa038c4b75ebc76dc855dd74f0da
Jimmy 1001 aad3b435b51404eeaad3b435b51404ee a9fdfa038c4b75ebc76dc855dd74f0da
Ava Evil 1002 aad3b435b51404eeaad3b435b51404ee 3e678b266d1d449e531e92bec33b6d27

C:\Users\admin123\Downloads\volatility3-develop\volatility3-develop>
```

Q3. User Credentials

- What are Ava Evil's and Jimmy's passwords?

Ava Evil's Hashed Password – aad3b435b51404eeaad3b435b51404eea

Jimmy's Hashed Password - aad3b435b51404eeaad3b435b51404eea

Q4. Login Activity

- What time did the following accounts log in?

o Ava Evil – 19:30

N/A	-	1916	MemCompression	-	2026-02-05 18:48:24.000000	UTC
2	-	5968	csrss.exe	-	2026-02-05 19:30:11.000000	UTC
2	-	2072	winlogon.exe	-	2026-02-05 19:30:11.000000	UTC
2	-	3056	dwm.exe	-	2026-02-05 19:30:12.000000	UTC
2	-	568	fontdrvhost.exe	-	2026-02-05 19:30:12.000000	UTC
2	-	6044	sihost.exe	WINDOWS11/Ava Evil	2026-02-05 19:30:20.000000	UTC
2	-	7984	svchost.exe	WINDOWS11/Ava Evil	2026-02-05 19:30:20.000000	UTC
2	-	7532	svchost.exe	-	2026-02-05 19:30:20.000000	UTC
2	-	5532	taskhostw.exe	-	2026-02-05 19:30:21.000000	UTC
2	-	7944	userinit.exe	-	2026-02-05 19:30:21.000000	UTC
2	Console	7672	explorer.exe	WINDOWS11/Ava Evil	2026-02-05 19:30:22.000000	UTC
2	-	2572	ShellHost.exe	-	2026-02-05 19:30:22.000000	UTC
2	-	8252	CrossDeviceRes	-	2026-02-05 19:30:23.000000	UTC
2	-	8820	svchost.exe	WINDOWS11/Ava Evil	2026-02-05 19:30:25.000000	UTC
2	-	9180	SearchHost.exe	-	2026-02-05 19:30:26.000000	UTC
2	-	9188	StartMenuExper	-	2026-02-05 19:30:26.000000	UTC
2	-	8508	svchost.exe	WINDOWS11/Ava Evil	2026-02-05 19:30:27.000000	UTC
2	-	9312	Widgets.exe	WINDOWS11/Ava Evil	2026-02-05 19:30:27.000000	UTC
2	-	9488	msedgewebview2	-	2026-02-05 19:30:28.000000	UTC
2	-	9532	msedgewebview2	-	2026-02-05 19:30:28.000000	UTC
2	-	9760	ShellExperienc	WINDOWS11/Ava Evil	2026-02-05 19:30:30.000000	UTC
2	-	9980	backgroundTask	WINDOWS11/Ava Evil	2026-02-05 19:30:30.000000	UTC
2	-	10060	msedgewebview2	-	2026-02-05 19:30:30.000000	UTC
2	-	10184	msedgewebview2	-	2026-02-05 19:30:30.000000	UTC
2	-	10216	msedgewebview2	-	2026-02-05 19:30:30.000000	UTC

o Jimmy – 19:31

Console	12756	OneDrive.exe	WINDOWS11/Ava Evil	2026-02-05 19:32:38.000000	UTC
-	12260	csrss.exe	-	2026-02-05 19:31:06.000000	UTC
-	11460	winlogon.exe	-	2026-02-05 19:31:06.000000	UTC
-	4092	dwm.exe	-	2026-02-05 19:31:07.000000	UTC
-	11916	fontdrvhost.exe	-	2026-02-05 19:31:07.000000	UTC
-	12304	sihost.exe	-	2026-02-05 19:31:14.000000	UTC
-	12324	svchost.exe	-	2026-02-05 19:31:14.000000	UTC
-	12364	svchost.exe	-	2026-02-05 19:31:15.000000	UTC
-	12436	taskhostw.exe	-	2026-02-05 19:31:15.000000	UTC
-	12672	userinit.exe	-	2026-02-05 19:31:16.000000	UTC
Console	12696	explorer.exe	WINDOWS11/Jimmy	2026-02-05 19:31:16.000000	UTC
-	12716	ShellHost.exe	-	2026-02-05 19:31:17.000000	UTC
-	12900	svchost.exe	-	2026-02-05 19:31:17.000000	UTC
-	13020	CrossDeviceRes	-	2026-02-05 19:31:18.000000	UTC
-	13272	SearchHost.exe	WINDOWS11/Jimmy	2026-02-05 19:31:19.000000	UTC
-	13280	StartMenuExper	-	2026-02-05 19:31:19.000000	UTC
-	2248	svchost.exe	-	2026-02-05 19:31:19.000000	UTC
-	13212	Widgets.exe	-	2026-02-05 19:31:20.000000	UTC
-	13404	msedgewebview2	-	2026-02-05 19:31:21.000000	UTC
-	13464	msedgewebview2	-	2026-02-05 19:31:22.000000	UTC
-	13776	backgroundTask	-	2026-02-05 19:31:23.000000	UTC

○ Admin123 – 18:48

-	10760	SearchProtocol	-	2026-02-05	19:30:37.000000	UTC
-	10956	SearchFilterHo	-	2026-02-05	19:30:38.000000	UTC
-	12224	WmiPrvSE.exe	WORKGROUP\WINDOWS11\$	2026-02-05	19:31:09.000000	UTC
-	16244	audiodg.exe	-	2026-02-05	19:32:24.000000	UTC
-	716	csrss.exe	-	2026-02-05	18:48:22.000000	UTC
-	764	winlogon.exe	-	2026-02-05	18:48:22.000000	UTC
-	968	fontdrvhost.ex	-	2026-02-05	18:48:22.000000	UTC
-	476	dwm.exe /SYSTEM	2026-02-05	18:48:23.000000	UTC	
-	4164	sihost.exe	-	2026-02-05	18:48:39.000000	UTC
-	4192	svchost.exe	-	2026-02-05	18:48:39.000000	UTC
-	4240	svchost.exe	-	2026-02-05	18:48:39.000000	UTC
-	4380	taskhostw.exe	-	2026-02-05	18:48:40.000000	UTC
-	4480	SMMonitor.exe	-	2026-02-05	18:48:40.000000	UTC
-	4548	userinit.exe	-	2026-02-05	18:48:40.000000	UTC
-	4580	explorer.exe	-	2026-02-05	18:48:40.000000	UTC
-	4624	ShellHost.exe	-	2026-02-05	18:48:40.000000	UTC
-	4848	svchost.exe	-	2026-02-05	18:48:41.000000	UTC
-	4980	CrossDeviceRes	WINDOWS11/admin123	2026-02-05	18:48:41.000000	UTC
-	5140	StartMenuExper	-	2026-02-05	18:48:42.000000	UTC
-	5152	SearchHost.exe	-	2026-02-05	18:48:42.000000	UTC
-	5316	Widgets.exe	-	2026-02-05	18:48:43.000000	UTC
-	5328	svchost.exe	-	2026-02-05	18:48:43.000000	UTC
-	5640	msedgewebview2	-	2026-02-05	18:48:44.000000	UTC
-	5668	msedgewebview2	-	2026-02-05	18:48:44.000000	UTC
-	5932	msedgewebview2	-	2026-02-05	18:48:45.000000	UTC
-	5960	msedgewebview2	-	2026-02-05	18:48:46.000000	UTC
-	6012	msedgewebview2	-	2026-02-05	18:48:46.000000	UTC
-	3384	msedgewebview2	-	2026-02-05	18:48:46.000000	UTC
-	6176	ShellExperienc	-	2026-02-05	18:48:46.000000	UTC
-	6764	WidgetService.	-	2026-02-05	18:48:48.000000	UTC
-	6352	ctfmon.exe	-	2026-02-05	18:48:50.000000	UTC
-	4684	SecurityHealth	-	2026-02-05	18:49:03.000000	UTC
-	7192	VBoxTray.exe	-	2026-02-05	18:49:03.000000	UTC
-	7476	RuntimeBroker.	-	2026-02-05	18:49:04.000000	UTC
-	7644	OneDrive.exe	-	2026-02-05	18:49:06.000000	UTC
-	7832	FileCoAuth.exe	-	2026-02-05	18:49:08.000000	UTC
-	7640	msedge.exe	-	2026-02-05	18:49:15.000000	UTC
-	4116	msedge.exe	-	2026-02-05	18:49:17.000000	UTC
-	7280	msedge.exe	-	2026-02-05	18:49:17.000000	UTC
-	7460	msedge.exe	-	2026-02-05	18:49:17.000000	UTC
-	7096	msedge.exe	-	2026-02-05	18:49:17.000000	UTC
-	6360	CodeMeterCC.ex	-	2026-02-05	18:49:21.000000	UTC

Q5. Running Notepad Instances

- How many Notepad instances were running in memory?

There were 2 instances of Notepad running. Initially, the pslist command was used to see if there were any instances of Notepad running. Then findstr Notepad was utilized to get a clean forensic

account of the instances and accurate count with dates and times.

```

Command Prompt
13272 1020 SearchHost.exe 0xaa0658d46080 37 - 3 False 2026-02-05 19:31:19.000000 UTC N/A Disabled
13280 1020 StartMenuExper 0xaa0658d44080 25 - 3 False 2026-02-05 19:31:19.000000 UTC N/A Disabled
2248 836 svchost.exe 0xaa0659eee080 9 - 3 False 2026-02-05 19:31:19.000000 UTC N/A Disabled
13212 1020 Widgets.exe 0xaa065645e080 16 - 3 False 2026-02-05 19:31:20.000000 UTC N/A Disabled
13404 13272 msedgewebview2 0xaa06582b7080 50 - 3 False 2026-02-05 19:31:21.000000 UTC N/A Disabled
13464 13404 msedgewebview2 0xaa0658640080 7 - 3 False 2026-02-05 19:31:22.000000 UTC N/A Disabled
13776 1020 backgroundTask 0xaa0658c53080 12 - 3 False 2026-02-05 19:31:23.000000 UTC N/A Disabled
13796 1132 taskhostw.exe 0xaa0658605080 0 - 2 False 2026-02-05 19:31:23.000000 UTC 2026-02-05 19:31:43.000000 UTC Disabled
13876 13404 msedgewebview2 0xaa06572c9080 17 - 3 False 2026-02-05 19:31:23.000000 UTC N/A Disabled
13936 13404 msedgewebview2 0xaa0658c05080 21 - 3 False 2026-02-05 19:31:23.000000 UTC N/A Disabled
14064 13404 msedgewebview2 0xaa06593c8080 9 - 3 False 2026-02-05 19:31:24.000000 UTC N/A Disabled
13200 13404 msedgewebview2 0xaa0658c26080 17 - 3 False 2026-02-05 19:31:27.000000 UTC N/A Disabled
13728 1020 WidgetService. 0xaa065660dd080 10 - 3 False 2026-02-05 19:31:28.000000 UTC N/A Disabled
9796 1212 ctfmon.exe 0xaa0658106080 11 - 3 False 2026-02-05 19:31:29.000000 UTC N/A Disabled
11332 7672 VBoxTray.exe 0xaa0657f4a080 8 - 2 False 2026-02-05 19:31:36.000000 UTC N/A Disabled
11732 12696 Notepad.exe 0xaa06560e4080 25 - 3 False 2026-02-05 19:31:40.000000 UTC N/A Disabled
9028 1020 RuntimeBroker. 0xaa0657793080 5 - 3 False 2026-02-05 19:31:43.000000 UTC N/A Disabled
12656 1020 RuntimeBroker. 0xaa0658a8a080 6 - 3 False 2026-02-05 19:31:43.000000 UTC N/A Disabled
11960 1132 taskhostw.exe 0xaa0658c51080 6 - 3 False 2026-02-05 19:31:43.000000 UTC N/A Disabled
11008 1020 smartscreen.ex 0xaa0658ca3080 9 - 3 False 2026-02-05 19:31:43.000000 UTC N/A Disabled
1412 12696 chrome.exe 0xaa06598ab080 48 - 3 False 2026-02-05 19:31:43.000000 UTC N/A Disabled
1144 1412 chrome.exe 0xaa0658135080 8 - 3 False 2026-02-05 19:31:43.000000 UTC N/A Disabled
14484 1412 chrome.exe 0xaa06593e7080 15 - 3 False 2026-02-05 19:31:45.000000 UTC N/A Disabled
14500 1412 chrome.exe 0xaa06593e2080 21 - 3 False 2026-02-05 19:31:45.000000 UTC N/A Disabled
14604 1412 chrome.exe 0xaa0658aae080 9 - 3 False 2026-02-05 19:31:46.000000 UTC N/A Disabled
14628 1132 taskhostw.exe 0xaa0659988080 7 - 2 False 2026-02-05 19:31:46.000000 UTC N/A Disabled
14812 1412 chrome.exe 0xaa06572c4080 14 - 3 False 2026-02-05 19:31:47.000000 UTC N/A Disabled
14820 1412 chrome.exe 0xaa0656077080 15 - 3 False 2026-02-05 19:31:47.000000 UTC N/A Disabled
14996 1412 chrome.exe 0xaa0658180c00 13 - 3 False 2026-02-05 19:31:51.000000 UTC N/A Disabled
15044 7028 msedge.exe 0xaa065863e080 13 - 2 False 2026-02-05 19:31:51.000000 UTC N/A Disabled
15136 1020 RuntimeBroker. 0xaa0655420080 8 - 2 False 2026-02-05 19:31:52.000000 UTC N/A Disabled
11076 5064 SearchProtocol 0xaa06591ef080 7 - 2 False 2026-02-05 19:31:55.000000 UTC N/A Disabled
16112 12696 SecurityHealth 0xaa0658107080 4 - 3 False 2026-02-05 19:31:56.000000 UTC N/A Disabled
8204 11460 LogonUI.exe 0xaa064dcd1080 12 - 3 False 2026-02-05 19:32:03.000000 UTC N/A Disabled
3580 7672 updatechecker. 0xaa0658aaa080 3 - 2 False 2026-02-05 19:32:07.000000 UTC N/A Disabled
15772 1020 RuntimeBroker. 0xaa065940b080 8 - 1 False 2026-02-05 19:32:17.000000 UTC N/A Disabled
16052 1020 RuntimeBroker. 0xaa0655de2080 10 - 1 False 2026-02-05 19:32:22.000000 UTC N/A Disabled
16244 2032 audiodg.exe 0xaa06524f4080 8 - 0 False 2026-02-05 19:32:24.000000 UTC N/A Disabled
16348 12696 VBoxTray.exe 0xaa0657b2f080 8 - 3 False 2026-02-05 19:32:27.000000 UTC N/A Disabled
16376 4580 FTK Imager.exe 0xaa0659e9f080 24 - 1 False 2026-02-05 19:32:27.000000 UTC N/A Disabled
12756 7672 OneDrive.exe 0xaa0654d4f080 38 - 2 False 2026-02-05 19:32:38.000000 UTC N/A Disabled
15400 1412 chrome.exe 0xaa0656ff0080 14 - 3 False 2026-02-05 19:32:48.000000 UTC N/A Disabled
186944185528776 0 0xaa0655c570c0 0 - 1439003560 True N/A N/A Disabled

C:\Users\admin123\Downloads\volatility3-develop\volatility3-develop>py -3.11 vol.py -f memdumpw6 windows.pslist | findstr Notepad
10524ess767200.0Notepad.exe 0xaa065732a080in23hed - 2 False 2026-02-05 19:30:45.000000 UTC N/A Disabled
11732 12696 Notepad.exe 0xaa06560e4080 25 - 3 False 2026-02-05 19:31:40.000000 UTC N/A Disabled

C:\Users\admin123\Downloads\volatility3-develop\volatility3-develop>

```

Q6. Network Connections

- What network connections were made by this computer?

Using Volatility3 to examine network activity, no active/recoverable network connections were identified in the image. Command was run successfully with no return, indicating the absence of network artifacts within the image.

```

Command Prompt
C:\Users\admin123\Downloads\volatility3-devel\volatility3-develop>py -3.11 vol.py -f memdumpw6 windows.netscan
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
Offset Proto LocalAddr LocalPort ForeignAddr ForeignPort State PID Owner Created
C:\Users\admin123\Downloads\volatility3-devel\volatility3-develop>py -3.11 vol.py -f memdumpw6 windows.netscan

```

Q7. Environment Variables

- What can you determine about the system's environment variables? Discuss three Environment variables.
 - 1) COMPUTERNAME identifies the hostname of the system as WINDOWS11. It's used by the OS and applications to reference the machine and is useful for system identification during analysis.

```
15400 chrome.exe 0x166c4fd7ac0 CommonProgramFiles C:\Program Files\Common Files
15400 chrome.exe 0x166c4fd7ac0 CommonProgramFiles(x86) C:\Program Files (x86)\Common Files
15400 chrome.exe 0x166c4fd7ac0 CommonProgramW6432 C:\Program Files\Common Files
15400 chrome.exe 0x166c4fd7ac0 COMPUTERNAME WINDOWS11
15400 chrome.exe 0x166c4fd7ac0 ComSpec C:\WINDOWS\system32\cmd.exe
15400 chrome.exe 0x166c4fd7ac0 DriverData C:\Windows\System32\Drivers\DriverData
15400 chrome.exe 0x166c4fd7ac0 EFC_12696_1262719628 1
```

- 2) USERNAME reveals the account under which process is executing. The presence of both user and system accounts indicates that there were interactive user sessions and core OS services occurring.

```
15400 chrome.exe 0x166c4fd7ac0 USERDOMAIN WINDOWS11
15400 chrome.exe 0x166c4fd7ac0 USERDOMAIN_ROAMINGPROFILE WINDOWS11
15400 chrome.exe 0x166c4fd7ac0 USERNAME Jimmy
15400 chrome.exe 0x166c4fd7ac0 USERPROFILE C:\Users\Jimmy
15400 chrome.exe 0x166c4fd7ac0 windir C:\WINDOWS
```

- 3) SystemRoot shows the location of the OS directory (C:\WINDOWS), It confirms the OS installation path and is useful for understanding system configuration and the file system structure.

```
dwm.exe 0x28723104a00 SystemDrive C:
dwm.exe 0x28723104a00 SystemRoot C:\WINDOWS
dwm.exe 0x28723104a00 TEMP C:\WINDOWS\TEMP
dwm.exe 0x28723104a00 TMP C:\WINDOWS\TEMP
```

Submission Requirements

- Submit a single document (PDF or DOCX) containing:
 - Answers to all questions
 - Supporting screenshots placed directly under each answer
- Clearly label each question and screenshot.

Academic Integrity Notice

This lab is an individual assignment. All work must be your own. Use of external answers or sharing results with others is not permitted and will result in a grade of 0 and will be reported to students' honor council.